Course Outline for CSE-432

Part A

1. Course Code: CSE-432

2. Course Title: Compiler Design (Sessional)

3. Course Type: Core Course

4.Level/ Term: Level: 4 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Dr. Mohammed Moshiul Hoque, Professor, Dept. of CSE, CUET
Ashim Dey, Lecturer, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 1.5 (3 hours of lab work per week)

9. Contact Hours: 3 hours of lab work per week

10. Total Marks: 150

11. Rational of the Course:

This course will cover different phases of compiler using high level programming techniques, working understanding of the process of lexical analysis, parsing and other compiler design aspects. This course is important to understand the fundamental role of different phases of compiler with its applications. This a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Sessional based on the following topics:

Introduction to compiling, Basic issues; Lexical analysis; Syntax analysis; Syntax-directed translation; Semantic analysis; Type checking; Run-time environments; Intermediate code generation; Code generation; Code optimization.

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12. Course Objectives:

(a) Understand and implement different phases of compiler using high level programming techniques.
(b) Demonstrate a working understanding of the process of lexical analysis, parsing and other compiler design aspects.
(c) Implement Lexical Analyzer using Lex tool & Syntax Analyzer or parser using YACC Tool

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

No.	Course Learning Outcomes (CLOs)	Bloom's Level (Optional)
CLO1	Understand the role of lexical analyzer, regular expression and transition diagrams in context of designing a compiler	C1, C4,A1,A4,P1
CLO2	Develop new code optimization techniques and apply it to improve the performance of a program in terms of speed & space	C5
CLO3	Apply the knowledge of lex tool & yacc tool to develop a scanner & parser.	C3

b) Mapping of CLOs with PLOs

No.	CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
1	CLO1	X											
2	CLO2			X									
3	CLO3					X							

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

Course Plan

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Introduction to Interpreter & compiler, phases of compilation, symbol table, compiler construction tools	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1

Week - 02	Implementation of lexical analyzer for given language with symbol table which ignores redundant spaces, tabs and new lines.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1
Week - 03	Implementation of tokenization of the lexemes into functions, operators, variable, numbers, keywords and strings.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1
Week - 04	Construct a program to test whether a given identifier is valid or not	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 05	Implementation of the operator precedence parsing of a given input	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 06	Implementation of a CFG to identify a given string is accepted / rejected.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 07	Implementation of a DFA to identify a given string is accepted / rejected.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation	Lab Performance Report	• CLO-1

		• Group Discussion		
Week - 08	Implementation of a NFA to identify a given string is accepted / rejected.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-1
Week - 09	Implementation of predicting parsing	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 10	Introduction to Lexical Analyzer using Lex tool & Syntax Analyzer or parser using YACC Tool	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-3
Week - 11	Construction of a LEX Program to convert the substring abc to ABC from the given input string; Implementation of a lexical analyzer using LEX and YACC	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-3
Week - 12	Lab Test			
Week - 13	Viva-voce Quiz			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Quizzes	15%
Viva-voce	15%
Class performance including reports	60%
Attendance	10%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%

b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedule if necessary.

Part D

16. Learning Materials

1) Recommended Readings

- "Compilers: Principles, Techniques, and Tools", Alfred V. Aho, Ravi Sethi, Jeffrey D. Ullman, (2nd Edition), Prentice hall, 2006. ISBN 0-321-48681-1

2) Supplementary Readings

- Kenneth C. Louden, Compiler Construction: Principles and Practice, Thomson Education, 1997
- Y. N. Srikant and Priti Shankar, The Compiler Design Handbook: Optimizations and Machine Code Generation, Second Edition, CRC Press.

3) Others